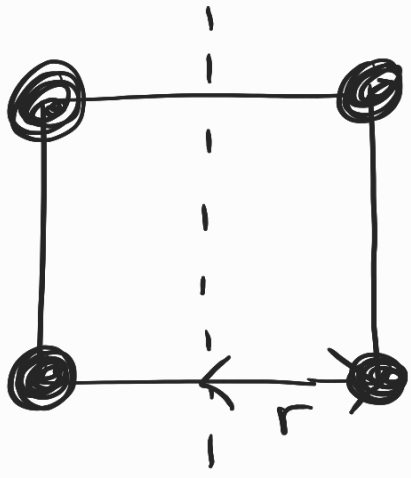
⑤ Παράδειγμα

$I = ?$ $\quad I = \sum m_i r_i^2$

$K_{rot} = ?$



$\Rightarrow r = \frac{2a}{2} = a$

$\Rightarrow I = 4 \cdot (ma^2)$

$K = \frac{1}{2} I \omega^2 = \frac{1}{2} \cdot 4ma^2\omega^2 = 2ma^2\omega^2$ //

⑦ Παράδειγμα

$I = ?$ $\qquad I = \int r^2 dm$ ①

$dm = \frac{dx}{L} \cdot M$ ② , $r^2 = x^2$

$\Rightarrow I = \int_0^L \frac{x^2 \cdot M}{L} dx$

$= \frac{M}{L} \left[\frac{x^3}{3} \right]_0^L = \frac{ML^3}{3L} = \frac{1}{3} ML^2$ //

(4) Παραβατόμα

$I_z = ?$ $\quad I_z = \sum m_i r_{zi}^2$

$= 3 \cdot 2^2 + 0 + 4(2\sqrt{2})^2 + 3 \cdot 2^2$

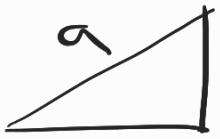
$r_1 = 2m$

$r_2 = 0m$

$r_3 = 2m$

$= 3 \cdot 4 + 4 \cdot 4 \cdot 2 + 3 \cdot 4$

$= 12 + 32 + 12 = 56 \ kgm^2$



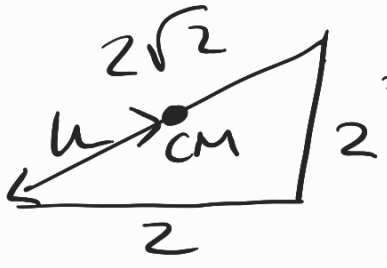
$\Rightarrow a^2 = 4 + 4 \Rightarrow a = \sqrt{8} = \sqrt{2 \cdot 4} = 2\sqrt{2} = r_4$

(15) $I_{CMz} = ?$

$I_{KM} + Mh^2 = I_z$

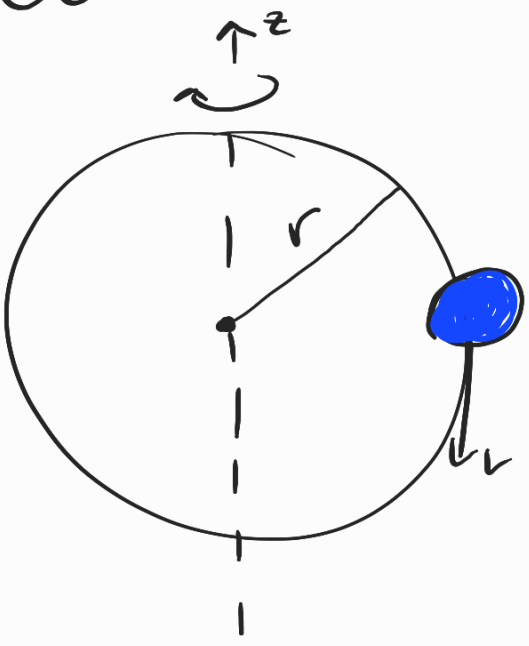
$\Rightarrow I_{KM} = I_z - Mh^2 = 56 - (2 \cdot 3 + 2 \cdot 4)(\sqrt{2})^2$

$= 28 \ kgm^2$



$\Rightarrow h = \frac{2\sqrt{2}}{2} = \sqrt{2}$

⑳ Παράδειγμα



$m = 2\,kg$

$v = 3.5\,m/s$

$r = 4\,m$

(a) $\vec{L} = ?$ $\quad \vec{L} = \vec{r} \times \vec{p} = r \cdot p \sin\varphi\, \hat{k}$

$= r \cdot m \cdot v \sin\varphi\, \hat{k}$

$r \perp v \perp p$

$\Rightarrow \varphi = 90°$

$= 4 \cdot 2(-3.5)\sin(90)\, \hat{k}$

$= -28\, \hat{k}\ \ kgm^2/s$

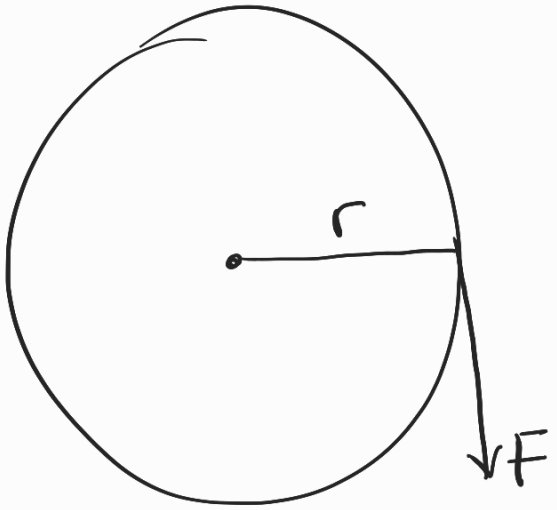
↳ right hand rule

(b) $I = ?$

$I = mr^2 = 2(4)^2 = 32\ kgm^2$

(c) $v = r\omega \Rightarrow \vec{\omega} = v/r = \frac{-3.5}{4} = -0.875\, \hat{k}\ rad/s$

(24) Παράδειγμα



$r = 2.2\,m$

$F = 260\,N$

$\Delta t = 12\,s = T$

(a) $a = ?$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{12} = \pi/6 \text{ rad/s}$$

$$x = u_0 t + \tfrac{1}{2} a t^2$$

$$\Rightarrow \vartheta = \cancel{\omega_0 t} + \frac{1}{2} a t^2$$

$\omega_0 = 0$

$$\Rightarrow \frac{1}{2} a t^2 = \Delta\vartheta \Rightarrow a = \frac{2\Delta\vartheta}{t^2} = \frac{2 \cdot 2\pi}{12^2} = 0.087 \text{ rad/s}^2$$

(b) $|\vec{\tau}| = ? \Rightarrow |\tau| = |r \times F|$

$r \perp F$

$$= rF\sin\varphi = (2.2)(260)\sin 90$$

$$= 572\,Nm$$

(c) $I = ?$

$$\tau = I a_{rad}$$

$$\Rightarrow I = \tau/a = \frac{572}{0.087} = 6.6 \times 10^3 \text{ kgm}^2$$

$\tau = F \cdot r$

$F = m \cdot a = m r a_{rad}$

$\Rightarrow \tau = m r^2 a_{rad}$

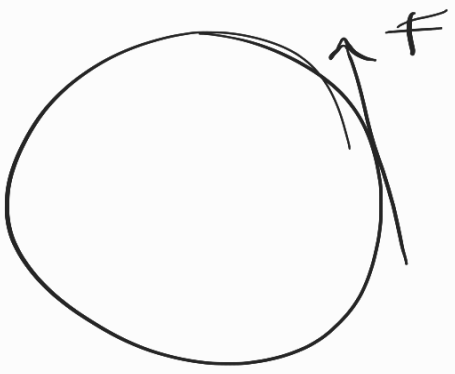
$= I a_{rad}$

$F_t = m \cdot a_t$

$a_t = r \cdot a_{rad}$

$\Rightarrow F_t = m r a_{rad}$

$\Rightarrow r F_t = m r^2 a_{rad}$



αλλά $\tau = rF$

$\Rightarrow \tau = m r^2 a$ $\quad I = mr^2$

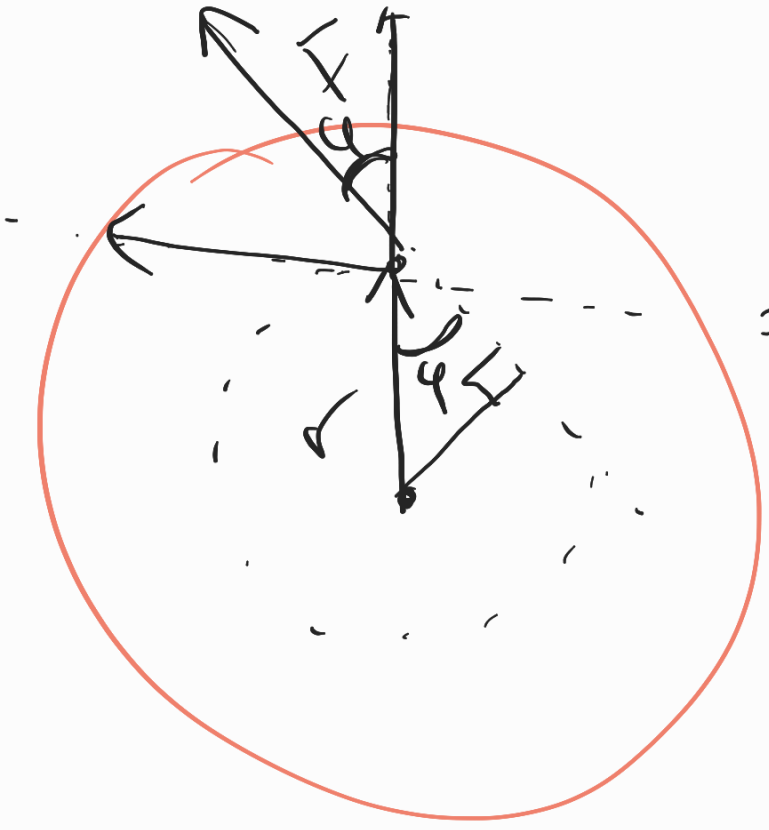
$\Rightarrow \tau = Ia$

ανάλογο της γραμμικής

F στην περιστροφική

κίνηση $\Rightarrow$ δύναμη

περιστροφής



sinφ

⇒ F sinφ